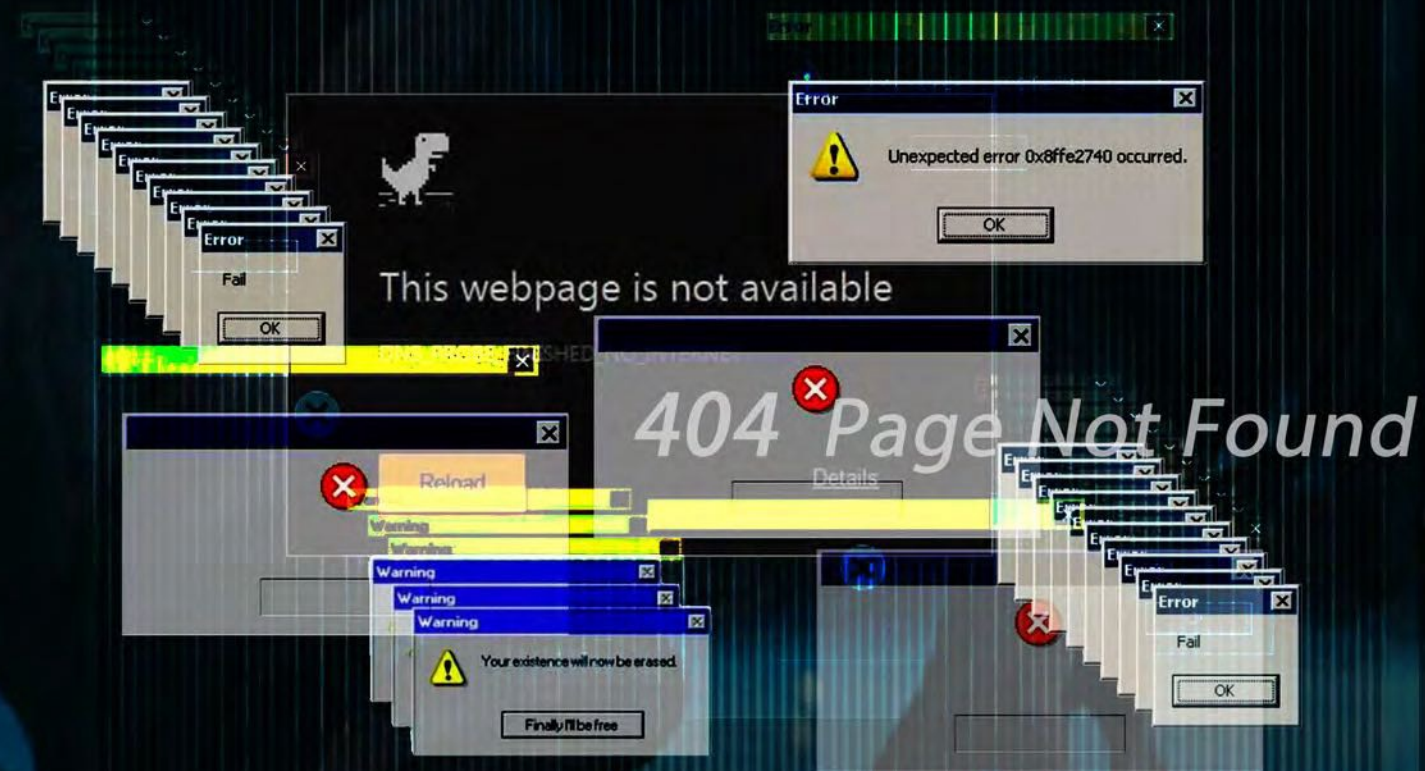
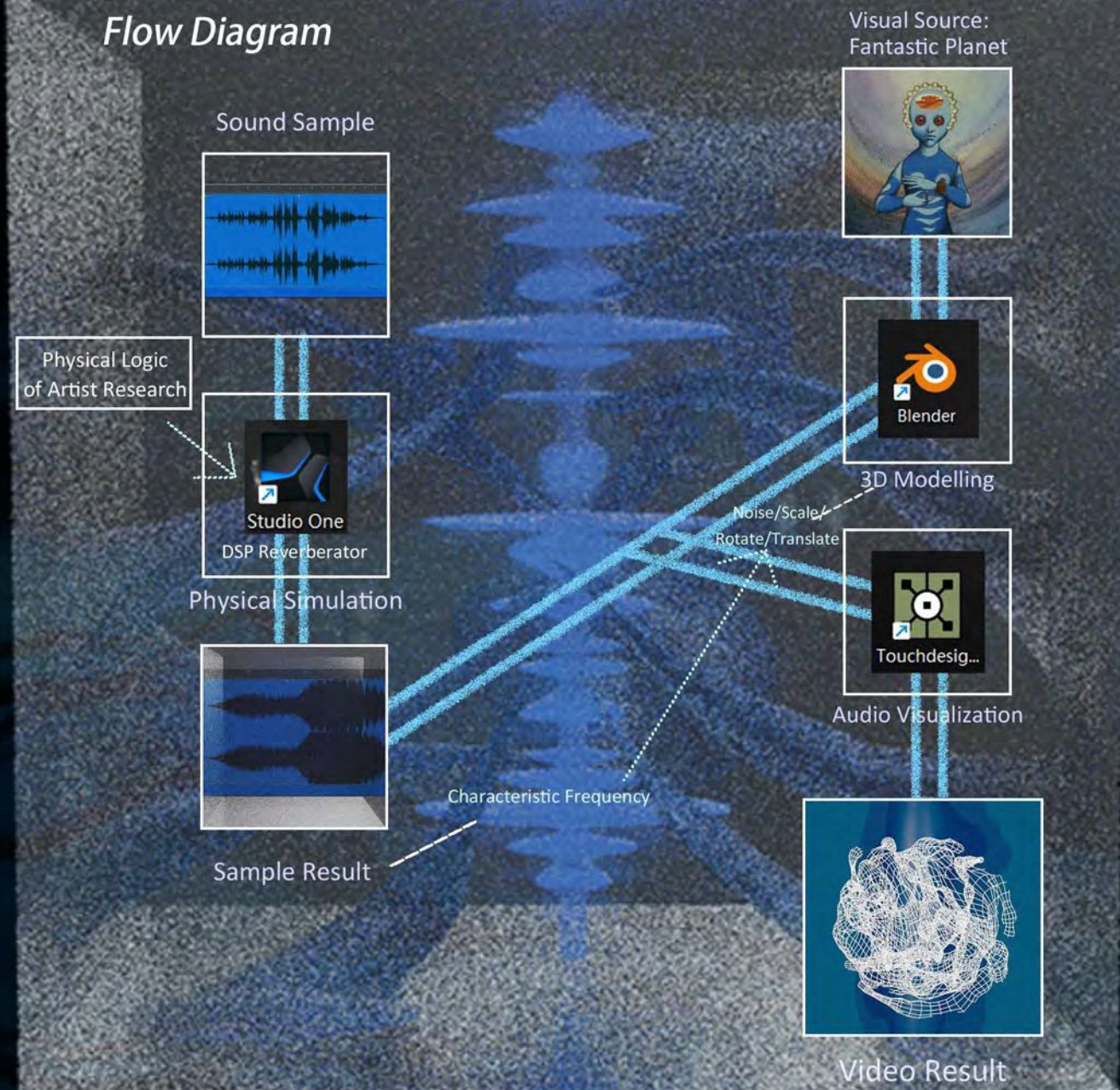




The Walls of China

Flow Diagram





Designer: Dominic Duan
Photographer: Tang
Model: Jiang

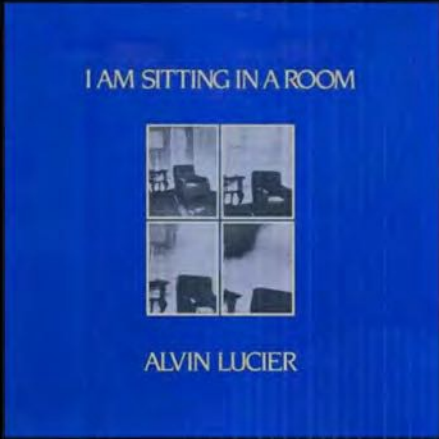






Designer: Dominic Duan
Photographer: Tang
Model: Jiang





Sound Experiment

<https://www.youtube.com/watch?v=zjiX4oc8e2U&list=WL&index=27&t>

I researched Alvin Lucier's 1969 sound experiment 'I Am Sitting in a Room.' Its physical logic inspired me: the human voice serves as the excitation source and the room as the excited system. In a closed room, sound repeats, amplifying the room's characteristic frequencies and absorbing others. Over time, only the room's characteristic frequencies remain.



Sound Sample Processing



I documented how the original audio's characteristic frequencies were gradually replaced by the system's frequencies.

Tools:

Studio One, Valhalla Reverb



1. Sample
2. Plugin: Reverb
3. Mixed down to a new track
4. Repeat 10 times



Initial characteristic frequencies



Final characteristic frequencies



Inspired by the movie "Fantastic Planet", which tells the story of the Oms who, after being enslaved and treated as toys by the Draags, learn advanced knowledge and rebel against their rulers, I created 3D models based on the wild Oms' life hidden in tree hollows during their oppression. I focused on their environment and colors to depict a society under strict control and limited information.

Process



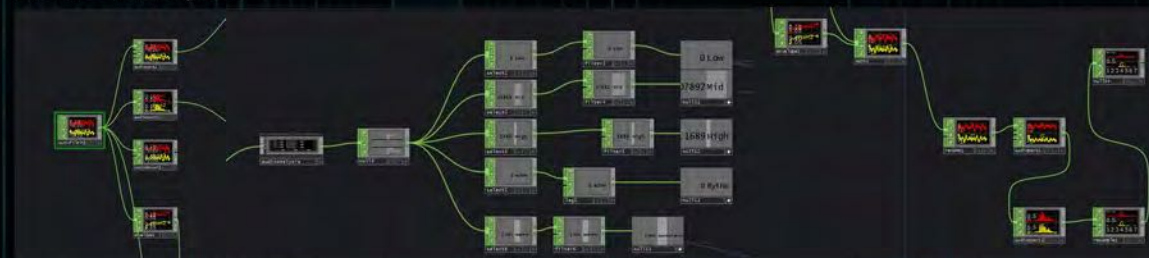
Three-view



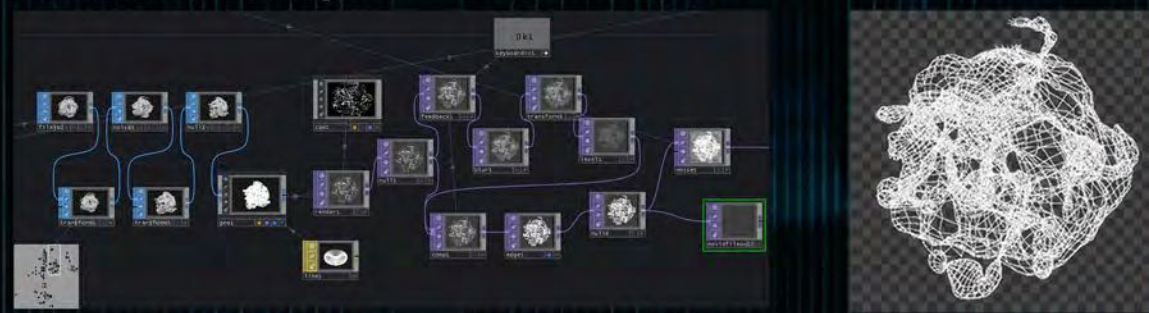
Then I used the audio from the sound experiment, utilizing its frequency, speed, and other parameters to control the morphological changes of the 3D model. This audio visualization illustrates the societal conditions under control.



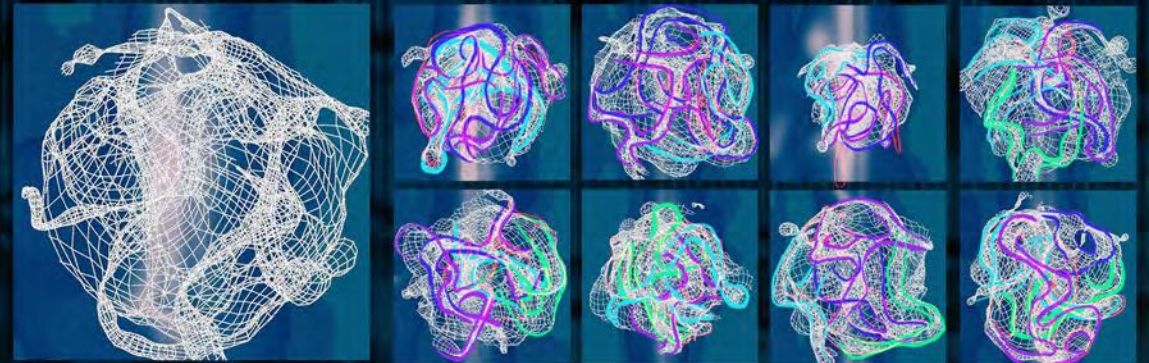
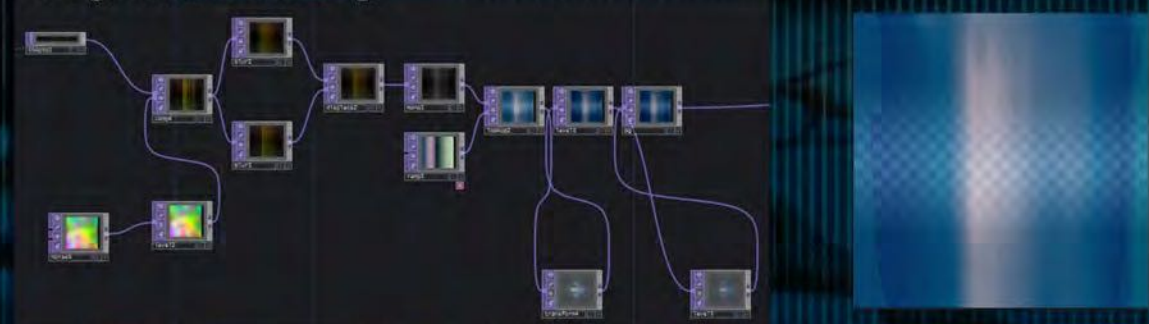
Audio Analysis



3D Model Processing



Background Processing



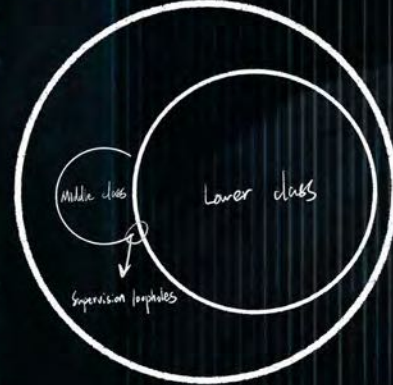
https://youtu.be/EBxTM3i_cPM



Pattern Experiment

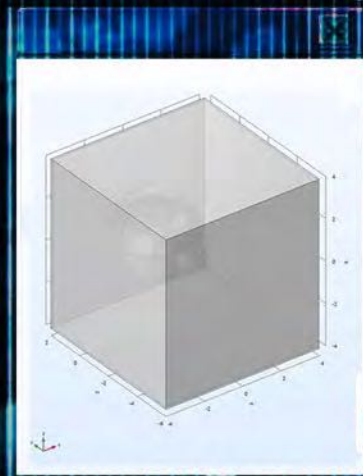
File Edit View Help

This sketch represents a society where the middle class is smaller than the lower class, and speech is heavily censored. Most voices are suppressed, with the lower class almost entirely silenced. However, some voices from the middle class slip through supervision loopholes,, offering effective solutions.

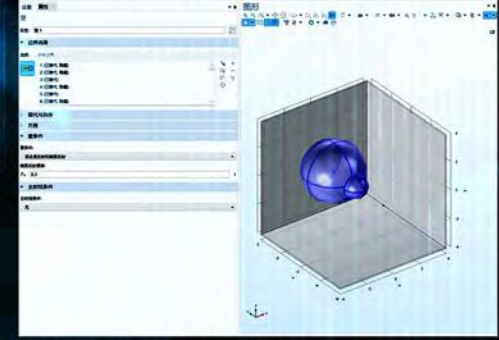


I modeled this concept and used COMSOL to simulate sound ray trajectories, capturing the leaked voices.

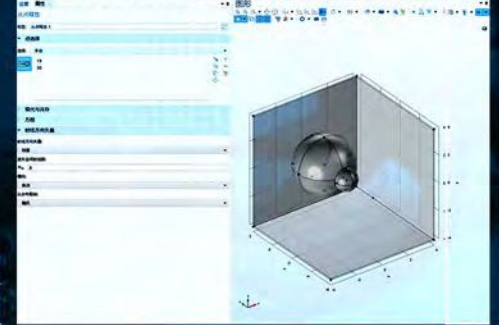
1. Modeling



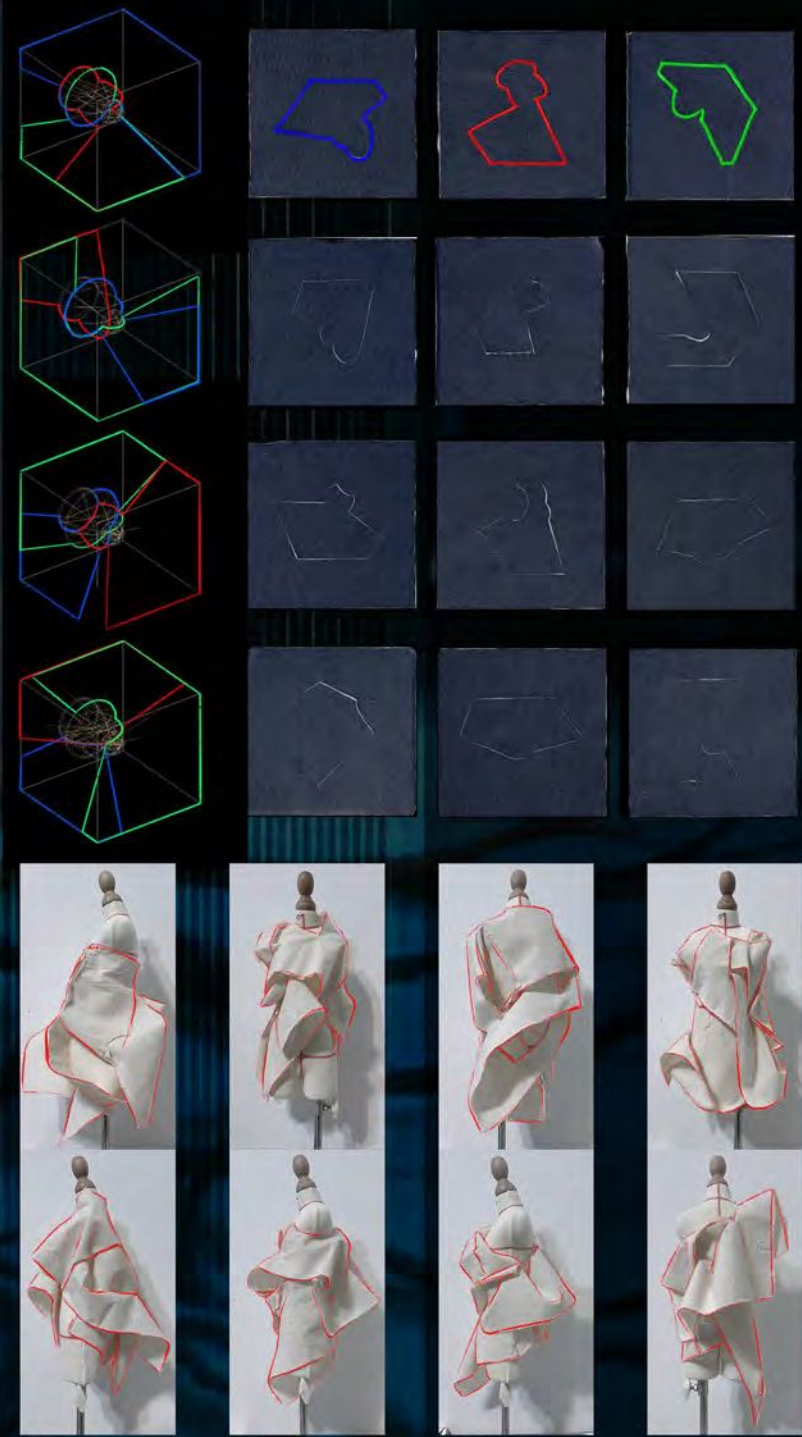
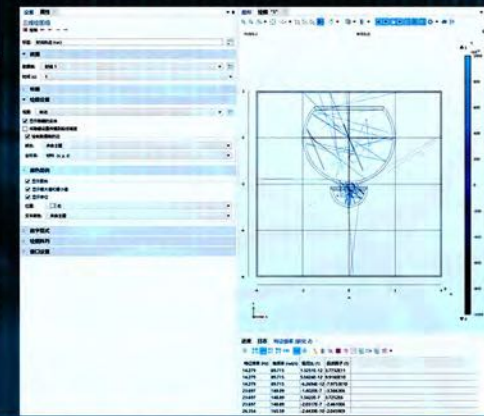
2. Set reflective surface parameters



3. Set sound source points



4. Calculate ray trajectories



3D Experiment

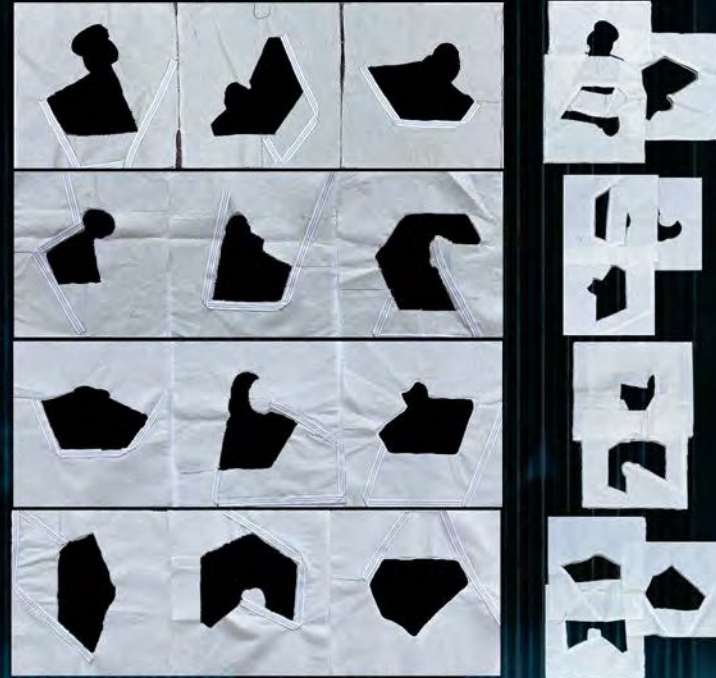


3D Development

Integrate the pattern pieces with the 3D elements



Pattern Deepening



1. Applying boning along the trajectory derived from the physics experiment.
2. Combine a set of patterns into one.



Draping and copy pattern.



Pattern Cutting & 3D Display



3D Scanning and Modeling



Fabric Development based on information processing

To get through the 'wall',I want to express how information flows on fabric and write code to simulate the process of information processing.

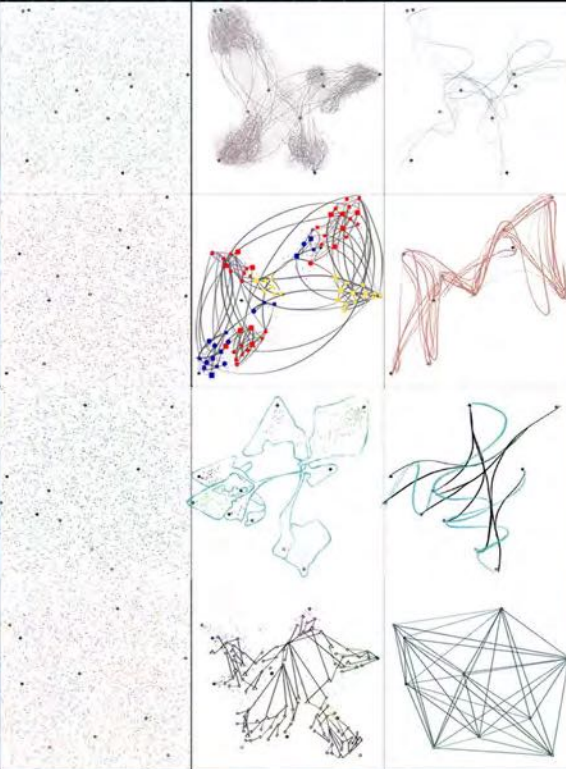
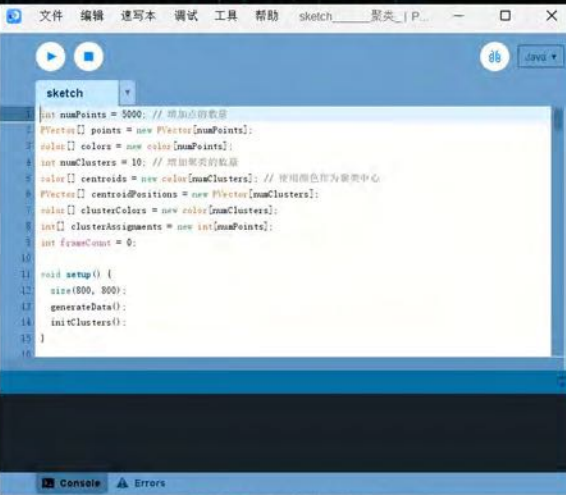
Method 1: Fabric Printing

The first approach is to use scenes from the movie as the primary ambient color, then apply a dot-matrix treatment.



Method 2: Fabric Modification

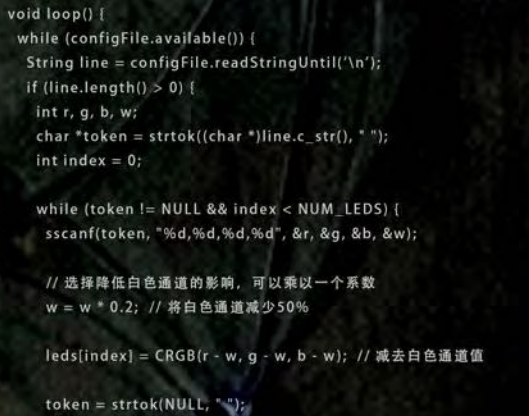
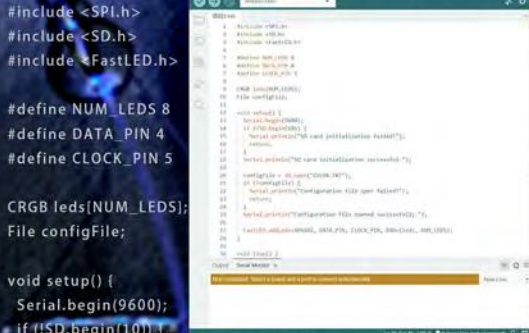
The second approach simulates the process of clustering and analyzing noise information.



Primarily using fiber optics, a medium for transmitting light signals, to modify the fabric.

Lighting Interaction

Materials List: Optical fiber, Arduino UNO, SK9822 5V, SD Card Module



Final Line-up



3D Clo & Fitting Process
- Look One



Look Three

